

Fullstack Developer

Interview Preparation Guide

200+ Questions · HTML · CSS · JavaScript · MERN · Gen AI · Auth · Deployment

HTML

CSS

JavaScript

React

Node/Express

MongoDB

Gen AI

Auth

Deploy

■ HTML

Basic

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|------|---|----|
| Q1. | What is HTML and what does it stand for? | [] |
| Q2. | What is the difference between HTML elements and HTML attributes? | [] |
| Q3. | What is the difference between div and span elements? | [] |
| Q4. | What are semantic HTML elements? Give 5 examples. | [] |
| Q5. | What is the difference between <section>, <article>, and <aside>? | [] |
| Q6. | What is the purpose of the declaration? | [] |
| Q7. | What is the difference between id and class attributes? | [] |
| Q8. | How do you add an image, link, and a table in HTML? | [] |
| Q9. | What is the difference between , , and <dl>? | [] |
| Q10. | What are void/self-closing elements? Give examples. | [] |
| Q11. | What is the difference between and , and <i>? | [] |

Intermediate

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|------|--|----|
| Q12. | What are HTML5 data attributes (data-*) and how are they used? | [] |
| Q13. | What is the difference between input[type=submit] and button elements? | [] |
| Q14. | What are meta tags? Name important ones for SEO and responsiveness. | [] |
| Q15. | How does the HTML form work? Explain action, method, enctype. | [] |
| Q16. | What is the difference between GET and POST in HTML forms? | [] |
| Q17. | What is the <canvas> element? How is it different from SVG? | [] |
| Q18. | What are ARIA attributes and why are they important for accessibility? | [] |
| Q19. | What is the difference between defer and async in <script> tags? | [] |
| Q20. | How does the browser render an HTML page step by step? | [] |

- Q21.** What is the DOM (Document Object Model)? []
- Q22.** What is the difference between innerHTML, innerText, and textContent? []

Advanced

- Q23.** What is the Critical Rendering Path and how can you optimize it? []
- Q24.** What are Web Components? Explain Custom Elements and Shadow DOM. []
- Q25.** How do you make an HTML page fully accessible (a11y)? []
- Q26.** What is the difference between localStorage, sessionStorage, and cookies? []
- Q27.** What is a Progressive Web App (PWA)? What HTML/meta tags does it need? []
- Q28.** What is the difference between display:none and visibility:hidden and how do they affect the DOM? []

■ CSS

Basic

- Q29.** What is CSS specificity? How is it calculated? []
- Q30.** What is the difference between margin and padding? []
- Q31.** What is the CSS box model? Explain content, padding, border, margin. []
- Q32.** What is the difference between relative, absolute, fixed, and sticky positioning? []
- Q33.** What is the difference between display: block, inline, and inline-block? []
- Q34.** How do CSS selectors work? Explain class, id, attribute, and pseudo selectors. []
- Q35.** What is the difference between em, rem, px, vh, and vw units? []
- Q36.** What is the z-index and how does stacking context work? []
- Q37.** What are CSS pseudo-classes (:hover, :nth-child, :focus) and pseudo-elements (::before, ::after)? []

Intermediate

- Q38.** How does Flexbox work? Explain justify-content, align-items, flex-wrap. []
- Q39.** How does CSS Grid work? Explain grid-template-columns, grid-area, gap. []
- Q40.** When would you use Flexbox vs CSS Grid? []
- Q41.** What is a CSS custom property (variable)? How do you use :root? []

Q42.	What is the difference between transition and animation in CSS?	[]
Q43.	What is CSS specificity conflict and how do you resolve it?	[]
Q44.	What is BEM naming convention and why is it useful?	[]
Q45.	How do you make a website responsive using media queries?	[]
Q46.	What is SCSS/SASS? What features does it add over plain CSS?	[]
Q47.	What is the difference between @import and @use in SCSS?	[]
Q48.	What is CSS glassmorphism and how do you implement it?	[]
Q49.	How do you center a div both horizontally and vertically?	[]

Advanced

Q50.	What is CSS containment (contain property) and how does it help performance?	[]
Q51.	What are CSS layers (@layer) and how do they help manage specificity?	[]
Q52.	How does will-change property work? When should you use it?	[]
Q53.	What is the difference between repaint and reflow in browser rendering?	[]
Q54.	How do CSS animations work on the GPU? What properties trigger GPU layers?	[]
Q55.	What is CSS-in-JS? What are the trade-offs vs traditional CSS/SCSS?	[]
Q56.	How do you implement a dark/light mode toggle using CSS variables only?	[]

■ JAVASCRIPT

Basic

Q57.	What is the difference between var, let, and const?	[]
Q58.	What is hoisting in JavaScript?	[]
Q59.	What is the difference between == and ===?	[]
Q60.	What are the 7 primitive data types in JavaScript?	[]
Q61.	What is the difference between null and undefined?	[]
Q62.	What is a closure in JavaScript? Give an example.	[]
Q63.	What is the difference between function declaration and function expression?	[]
Q64.	What are arrow functions? How are they different from regular functions?	[]
Q65.	What is the difference between forEach, map, filter, and reduce?	[]

Q66.	What is the difference between splice and slice?	[]
Q67.	How does the spread operator (...) work?	[]
Q68.	What is destructuring in JavaScript?	[]
Q69.	What are template literals (backticks)?	[]

Intermediate

Q70.	What is the JavaScript event loop? How does it work?	[]
Q71.	What is the call stack vs the task queue vs the microtask queue?	[]
Q72.	What is a Promise? How is it different from a callback?	[]
Q73.	What is async/await and how does it simplify Promises?	[]
Q74.	What is Promise.all(), Promise.race(), Promise.allSettled()?	[]
Q75.	What is the difference between shallow copy and deep copy?	[]
Q76.	What is prototypal inheritance in JavaScript?	[]
Q77.	What is the difference between __proto__ and prototype?	[]
Q78.	What is the 'this' keyword? How does it behave in arrow vs regular functions?	[]
Q79.	What are higher-order functions? Give examples.	[]
Q80.	What is event bubbling and event capturing?	[]
Q81.	What is event delegation and why is it useful?	[]
Q82.	What is debounce vs throttle? When do you use each?	[]
Q83.	What is the difference between synchronous and asynchronous code?	[]
Q84.	What are WeakMap and WeakSet used for?	[]
Q85.	What is a generator function (function*)? How does yield work?	[]
Q86.	What is the module system in JavaScript (ES Modules vs CommonJS)?	[]
Q87.	What is optional chaining (?.) and nullish coalescing (??)?	[]

Advanced

Q88.	How does garbage collection work in JavaScript?	[]
Q89.	What is a memory leak in JavaScript? How do you detect and prevent it?	[]
Q90.	What are Proxy and Reflect in JavaScript? Give use cases.	[]
Q91.	What is the difference between compile-time and runtime errors?	[]

Q92.	How does JavaScript handle concurrency without threads?	[]
Q93.	What are Symbol and its use cases?	[]
Q94.	What is tail call optimization?	[]
Q95.	How does tree-shaking work in bundlers like Vite/Webpack?	[]
Q96.	What is the temporal dead zone (TDZ)?	[]
Q97.	Explain memoization and implement it in JavaScript.	[]

■■ REACT.JS

Basic

Q98.	What is React and why do we use it over plain JavaScript?	[]
Q99.	What is JSX? How does the browser understand it?	[]
Q100	What is the difference between Props and State?	[]
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Q101	What does useState hook do? Give an example.	[]
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Q102	When and why do we use useEffect?	[]
.		
Q103	How do you handle events in React?	[]
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Q104	How do you do conditional rendering in React?	[]
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Q105	Why is the key prop important when rendering lists?	[]
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Q106	What is the difference between a controlled and uncontrolled component?	[]
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Intermediate

Q107	What is useContext? How does it solve prop drilling?	[]
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Q108	What is useRef and when do you use it instead of useState?	[]
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Q109	What is the difference between useMemo and useCallback?	[]
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Q110	What are custom hooks? Build a useFetch() example.	[]
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Q111	How does React Router DOM work? How do you create protected routes?	[]
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Q112	What is lazy loading and Suspense in React?	[]
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Q113	What is the difference between useEffect with [], with deps, and with no array?	[]
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Q114	How do you share logic between components without prop drilling?	[]
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Q115	What are React portals and when would you use them?	[]
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Q116	What is the difference between client-side and server-side rendering in React?	[]
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Q117	How do you handle forms in React (controlled vs libraries like React Hook Form)?	[]
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Advanced

Q118	What is the Virtual DOM and how does reconciliation work?	[]
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Q119	What is React Fiber and how did it improve the rendering engine?	[]
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Q120	When would you use useReducer instead of useState?	[]
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Q121	What are error boundaries and how do you implement one?	[]
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Q122	What is React.memo and when does it actually help?	[]
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Q123	When would you choose Zustand or Redux over Context?	[]
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Q124	What are React Server Components? How are they different from Client Components?	[]
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Q125	How do you avoid stale closures in useEffect?	[]
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Q126	How do you profile and optimize a slow React app?	[]
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■ NODE.JS

Basic

Q127	What is Node.js and how is it different from browser JavaScript?	[]
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Q128	Node.js is single-threaded — how is it still fast and non-blocking?	[]
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Q129	What is the event loop in Node.js?	[]
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Q130	What is npm? What is the role of package.json?	[]
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Q131	What is the difference between require() and import?	[]
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Q132	What is the difference between callback, Promise, and async/await?	[]
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Intermediate

Q133	What is process.env and how do you use dotenv?	[]
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Q134	What are Node.js streams? When are they useful?	[]
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Q135	How do you hash passwords using bcrypt in Node.js?	[]
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Q136	What is JWT? Explain access token vs refresh token.	[]
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Q137	What is the difference between storing auth in cookies vs localStorage?	[]
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Q138	How does Node.js clustering work?	[]
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Q139	What is the difference between setImmediate, process.nextTick, and setTimeout?	[]
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Advanced

Q140	How do you detect and fix memory leaks in a Node.js application?	[]
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Q141	How does Socket.io work with Node.js for real-time communication?	[]
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Q142	What is PM2 and how do you use it in production?	[]
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Q143	How do you handle uncaught exceptions and unhandled promise rejections?	[]
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■ EXPRESS.JS

Basic

Q144	What is Express.js and how is it different from plain Node.js?	[]
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Q145	What is middleware in Express and how does it work?	[]
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Q146	What is the difference between <code>app.use()</code> and <code>app.get()</code> ?	[]
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Q147	What is the difference between <code>req.params</code> , <code>req.query</code> , and <code>req.body</code> ?	[]
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Q148	What is a REST API? Explain HTTP methods: GET, POST, PUT, PATCH, DELETE.	[]
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Q149	How do you create a router in Express?	[]
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Intermediate

Q150	How do you create an error handling middleware in Express?	[]
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Q151	What is CORS? How do you handle it in Express?	[]
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Q152	What does <code>express.json()</code> middleware do?	[]
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Q153	How do you implement rate limiting in Express?	[]
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Q154	What is the MVC pattern and how do you follow it in Express?	[]
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Q155	How do you handle file uploads using Multer?	[]
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Q156	What is the difference between PUT and PATCH?	[]
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Advanced

Q157	In Express 5, async errors are automatically caught — explain how and why.	[]
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Q158	How do you design a scalable middleware chain for large applications?	[]
.		
Q159	How do you implement caching in an Express API?	[]
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Q160	How do you write integration tests for Express routes?	[]
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Basic

- Q161** What is MongoDB and how is it different from SQL databases? []
.
- Q162** What is the difference between a Document and a Collection? []
.
- Q163** How do you perform CRUD operations in MongoDB? []
.
- Q164** What is an ObjectId and how is it automatically generated? []
.
- Q165** Does MongoDB enforce a schema? Explain. []
.
- Q166** What is the difference between find() and findOne()? []
.
- Q167** What do \$set, \$push, and \$pull operators do? []
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Intermediate

- Q168** What is Mongoose and why do we use it? []
.
- Q169** How do you define a Schema and Model in Mongoose? []
.
- Q170** What is populate() in Mongoose and when do you use it? []
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- Q171** What is indexing in MongoDB and how does it affect performance? []
.
- Q172** What is the Aggregation Pipeline? Explain \$match, \$group, \$project. []
.
- Q173** How do you model relationships (one-to-many, many-to-many) in MongoDB? []
.
- Q174** When should you use MongoDB transactions? []
.

Advanced

- Q175** What is MongoDB Atlas? How is it different from a local MongoDB instance? []
.
- Q176** What is a compound index and when should you use one? []
.
- Q177** Explain sharding and replication in MongoDB. []
.

Q178	What is a sparse index and what is its use case?	[]
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Q179	How do you implement full-text search using a text index?	[]
.		
Q180	What is the difference between embedding and referencing documents?	[]
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■ GEN AI / LLMs

Basic

Q181	What is a Large Language Model (LLM)?	[]
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Q182	What is prompt engineering? How do you write a good prompt?	[]
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Q183	What is a token in the context of LLMs?	[]
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Q184	How do you call the OpenAI / Gemini API from Node.js?	[]
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Q185	What does the temperature parameter control in an LLM response?	[]
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Q186	What is the difference between a system prompt and a user prompt?	[]
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Q187	What are the differences between GPT-4, Gemini, and Mistral?	[]
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Intermediate

Q188	What is RAG (Retrieval Augmented Generation) and why is it used?	[]
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Q189	What are vector embeddings? How does semantic search work?	[]
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Q190	What is LangChain and what problem does it solve?	[]
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Q191	What is streaming in LLM APIs? How do you implement it with Socket.io?	[]
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Q192	How do you build an auto-tagging feature using an LLM?	[]
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Q193	What is function calling / tool use in LLMs?	[]
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Q194	What is the Tavily API and how is it used for web search in AI agents?	[]
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Q195	What is an AI agent architecture?	[]
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Q196	What is the difference between zero-shot, one-shot, and few-shot prompting?	[]
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Advanced

Q197	What is a knowledge graph and how is it used with LLMs?	[]
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Q198	How do you implement topic clustering using an LLM?	[]
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Q199	When should you use Fine-tuning vs RAG? Compare trade-offs.	[]
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Q200	What is LLM hallucination and how do you reduce it?	[]
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Q201	What is a multi-agent system and when would you use it?	[]
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Q202	What is the difference between semantic search and keyword search?	[]
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Q203	What are vector databases like Pinecone or ChromaDB and when do you need them?	[]
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Q204	What is chain-of-thought prompting and how does it improve reasoning?	[]
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Q205	How do you evaluate the output quality of an LLM in production?	[]
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■ AUTH + SECURITY

All Levels

Q206	How does JWT authentication work? Explain step by step.	[]
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Q207	Explain the Google OAuth 2.0 authorization flow.	[]
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Q208	What is an httpOnly cookie? Why is it more secure than localStorage for tokens?	[]
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Q209	What is a CSRF attack and how do you prevent it?	[]
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Q210	What is SQL Injection and NoSQL Injection? How do you prevent each?	[]
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Q211	What does Helmet.js do in an Express application?	[]
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Q212	What is the refresh token rotation strategy?	[]
Q213	How do you implement a secure password reset flow?	[]
Q214	What is the difference between authentication and authorization?	[]
Q215	What is bcrypt and why is it used instead of MD5 or SHA?	[]
Q216	What is XSS (Cross-Site Scripting) and how do you prevent it in React apps?	[]
Q217	What is HTTPS and why should every production app use it?	[]

■ DEPLOYMENT + DEVOPS

All Levels

Q218	How do you deploy a MERN stack application on Render?	[]
Q219	How do you connect a custom domain from GoDaddy to Render?	[]
Q220	How do you manage environment variables in production securely?	[]
Q221	What is Cloudinary and how do you handle file uploads with it?	[]
Q222	What is a Vite base path misconfiguration and how do you fix it?	[]
Q223	How do you resolve CORS errors that appear only in production?	[]
Q224	How do you send transactional emails using the Resend API?	[]
Q225	How do you secure a MongoDB Atlas connection string in production?	[]
Q226	What is the difference between vertical and horizontal scaling?	[]
Q227	What is a CI/CD pipeline? How does it work?	[]
Q228	What is Docker? Why is containerization useful for Node.js apps?	[]
Q229	What is the difference between a monolith and a microservices architecture?	[]

■ HR + BEHAVIORAL

Must Prepare

Q230	Tell me about your most challenging project and a problem you solved.	[]
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Q231	Why did you choose the MERN stack? Would another stack have been better?	[]
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Q232	Describe a time you got stuck on a feature and how you handled it.	[]
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Q233	Explain one of your AI features to someone non-technical.	[]
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Q234	Where do you see yourself in 5 years?	[]
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Q235	How do you handle disagreements in a team?	[]
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Q236	Tell me about a project that taught you what NOT to do.	[]
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Q237	How do you stay updated with new technologies?	[]
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Q238	Describe your code review process.	[]
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Q239	How do you estimate how long a feature will take to build?	[]
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Q240	Have you ever missed a deadline? What happened and what did you learn?	[]
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Q241	What is one technical concept you learned recently and found interesting?	[]
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